foodZILLA: The Beast of Healthy Recipes

Project documentation for the Afro-Asian Forum for Innovation and Technology (AFFIAT), Aswan Finals 2026.



> Every number in this document was checked against the code (`index.html`, `style.css`, `project.js`, `app.js`). If you change the app, update this file and run `python docs/build.py` to rebuild the PDF, the poster, the slides and the diagrams.

1. Project Overview | نظرة عامة على المشروع

Field	Value
Project Title	foodZILLA: The Beast of Healthy Recipes
Project Type	Static website (HTML, CSS and JavaScript, no server)
Description	foodZILLA is a recipe website with a Godzilla movie look. It holds 31 recipes in five nutrition groups. Each recipe opens in a popup with a YouTube cooking video, its ingredients, numbered steps, calories and price, and can be added to a shopping cart with a delivery fee.
Target Users	Teens and young adults who think healthy food is boring, university students who want cheap and quick meals, athletes who count protein and calories, and families who cook together at home.
Team	Abdel Wahab
Source code	https://github.com/mr-basem/Foodzilla

Field	Value
Live app	https://mr-basem.github.io/Foodzilla/

Tech Stack | التقنيات المستخدمة

Category	Technology	Purpose
Structure	HTML5 (`index.html`)	The navbar, the Godzilla header, the search box, the category buttons, the recipe popup, the Sign in panel and the cart drawer
Styling	CSS3 (`style.css`, 1,045 lines)	The fire background, the lightning flash, the atomic fire beam, the shaking title, the city skyline, card grid and a phone layout under 520 px
Data	JavaScript object (`project.js`)	The 31 recipes, grouped by category
Logic	JavaScript (`app.js`)	Showing recipes, category filter, search, popup, cart, delivery fee and saved delivery details
Storage	Browser `localStorage`	The customer's delivery details, under the key `foodzillaCustomer`
Video	YouTube embeds	One cooking video per recipe, played in an `<iframe>`
Hosting	GitHub Pages	Serves the files at mr-basem.github.io/Foodzilla

Dependencies | المكتبات

Package	Version	Purpose
None	–	The app uses no libraries, frameworks or build tools. Any modern browser opens `index.html` directly.
YouTube	Online service	The recipe videos need an internet connection. Everything else works offline.

2. Problem Statement | المشكلة

Many teens and young adults believe that healthy food is boring or hard to cook, so they choose fast food instead. When they do look for a healthy recipe, the pieces are scattered: the steps are on one website, the video on another, and the calories are often missing.

Why does it matter? | لماذا هذا مهم؟

Habits formed in the teenage years tend to last. A young person who never learns a few simple, balanced meals keeps relying on fast food, which is high in fat, sugar and salt. Knowing how many calories a meal has, and what group of nutrients it belongs to, helps them make better choices.

How is it currently solved? | كيف تُحل المشكلة حاليًا؟

Today people search cooking websites, scroll through social media videos, or order from delivery apps. Recipe sites rarely show calories next to a video, social media videos rarely list exact ingredients, and delivery apps do not teach you how to cook. foodZILLA puts the recipe, the video, the calories and a cart in one place, and gives it a bold, game-like look that young people want to explore.

3. Technical Architecture | البنية التقنية

Project Structure | هيكل المشروع

```
Foodzilla/
├── index.html        # The page and its popup, Sign in panel and cart
├── style.css         # The Godzilla theme, animations and phone layout
├── project.js        # The recipe data: 31 recipes in 5 lists
├── app.js            # Show, filter, search, popup, cart, delivery
├── download (N).jpg  # Recipe photos (31 used by project.js)
├── template.html     # Early Arabic version (not linked)
├── hi.html, hi.css   # A small practice page (not linked)
├── README.md
└── docs/             # Documentation, poster, slides, diagrams
```

Architecture and Data Flow | تدفق البيانات

```
Browser opens index.html
    → loads style.css (the look)
    → loads project.js (the "recipes" object, 5 lists)
    → loads app.js
        → displayRecipes('all') draws a card for every recipe
        → updateAccountButton() reads saved delivery details
        → renderCart() draws the empty cart

Category button → displayRecipes(category)
Search box      → keeps recipes whose name OR an ingredient matches
Recipe card     → showRecipeDetails(): video, kcal, price, steps
Add to Cart     → addToCart() → renderCart(): subtotal + 50 EGP
Buy Now (cart)  → checkoutCart(): empty? no address? → placed
```

There is no server and no database. The recipes are written in `project.js`, the cart lives in the page's memory, and the delivery details are kept in the browser's `localStorage`.



Key Files | الملفات الأساسية

File	Purpose
`index.html`	All the page's parts. The Godzilla silhouette is drawn in an inline SVG inside the header. The popup, Sign in panel and cart drawer start hidden and are shown by adding the `active` class.
`style.css`	The cinematic look: `fireBackgroundShift`, `movieLightning`, `seismicRumble`, `fireBlastCycle` and `screenShake` animations, the skyline cut out with `clip-path`, and the `@media (max-width: 520px)` phone layout.
`project.js`	One object called `recipes` with five lists. Each recipe has `name`, `icon` (photo file), `price`, `ingredients`, `calories`, `youtubeUrl` and `steps`.
`app.js`	430 lines in two parts. Part one: `displayRecipes`, `showRecipeDetails`, `closeModal`, the category and search listeners. Part two: the Sign in form, delivery details, `renderCart`, `addToCart`, `buyNow`, `changeQty`, `removeFromCart`, `clearCart`, `checkoutCart` and `toggleCart`.

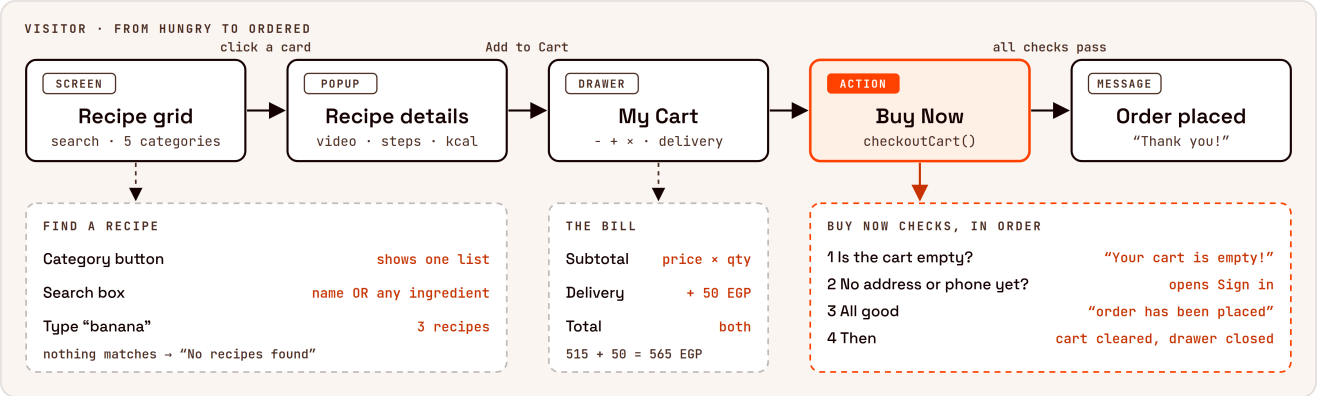
4. Features & User Flow | المميزات ورحلة المستخدم

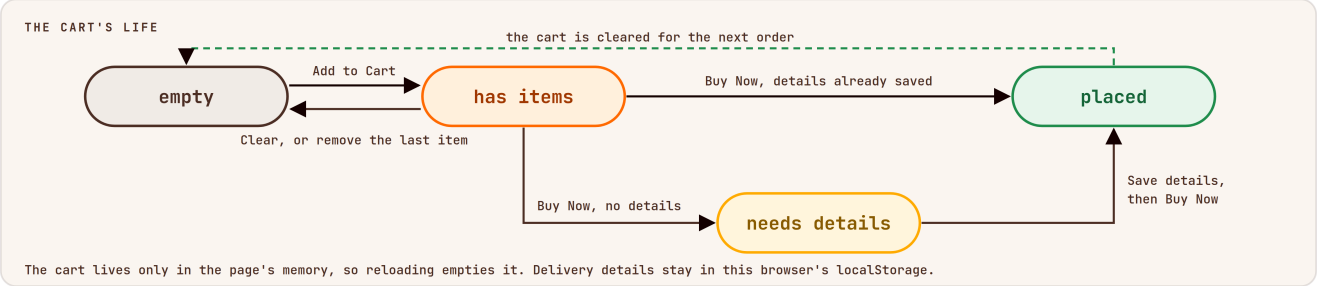
Feature List | قائمة المميزات

#	Feature	Description	Status
1	Cinematic header	Godzilla silhouette, a fire beam across the title, lightning flashes, a shaking title and a city skyline, all in pure CSS.	Complete

#	Feature	Description	Status
2	Nutrition groups	Six buttons: all, Protein (4), Carbohydrates (7), Sweets (8), Fruits (7) and Dairy (5).	Complete
3	Instant search	Searches each recipe's name and every ingredient as you type. Shows "No recipes found" when nothing matches.	Complete
4	Recipe cards	Photo, name, calories, price, and "Add to Cart" and "Buy Now" buttons.	Complete
5	Recipe popup	YouTube video, calories, price, ingredient list and numbered steps. Closing it stops the video.	Complete
6	Sign in for delivery	Name, governorate, city, address, phone (10 or 11 digits) and a password of 8 to 16 characters. A show/hide eye button for the password.	Complete. The password is checked but never saved or used to log in.
7	Cart drawer	− / + / × for each meal, subtotal, a 50 EGP delivery fee and the total. A badge on the Cart button counts the items.	Complete. Emptied when the page reloads.
8	Delivery details in the cart	Shows the saved name, address and phone, with an Edit form inside the cart.	Complete
9	Checkout	Buy Now warns if the cart is empty, opens Sign in if there is no address or phone, and otherwise says the order has been placed and clears the cart.	Demo only. The order is not sent anywhere.
10	Phone layout	Cards, buttons, popups and the cart resize for screens under 520 px wide.	Complete

User Flow | رحلة المستخدم






```
Home page (index.html)
    → pick a category button, or type in the search box
    → click a recipe card → popup (video, ingredients, steps, kcal)
        → Add to Cart  (shows an "added to cart!" message)
        → Buy Now      (adds it and opens the cart)
    → Cart button → My Cart drawer
        → - / + / × to change amounts
        → Buy Now
            → cart empty?           "Your cart is empty!"
            → no address or phone?  the Sign in panel opens
            → otherwise             "Your order has been placed."
                                    and the cart is cleared
```

Screen Reference | الشاشات

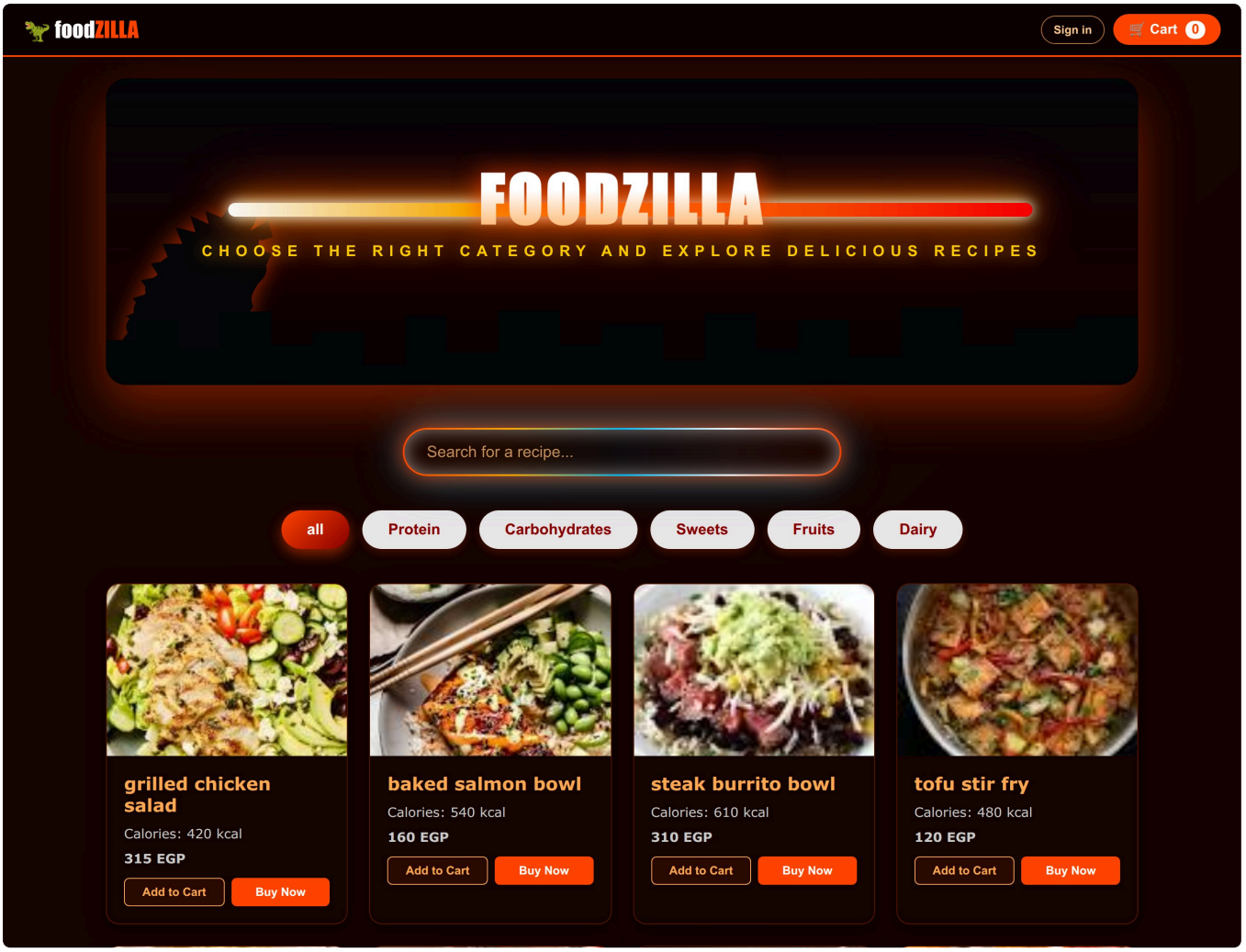
The app is a single page. These are its parts:

Part	Where in the code	Description
Navbar	`index.html` `.top-nav`	"🦖 foodZILLA" logo, the Sign in button (shows your name once saved) and the Cart button with a count badge.
Header	`index.html` `<header>`	The animated Godzilla scene with the title "FOODZILLA" and "Choose the right category and explore delicious recipes".
Search and categories	`#searchBar`, `#categories`	The search box and the six group buttons.
Recipe grid	`#recipesGrid`	One card per recipe, four per row on a laptop, one per row on a phone.

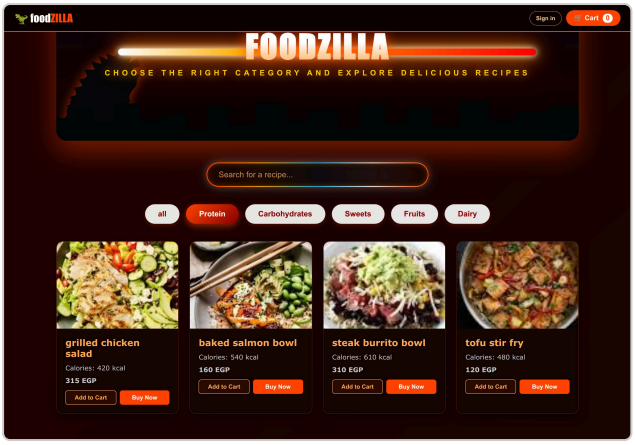
Part	Where in the code	Description
Recipe popup	`#recipeModal`	The full recipe with its video. Closes with × or by clicking outside.
Sign in panel	`#accountOverlay`	The delivery details form.
Cart drawer	`#cartDrawer`	Slides in from the right with delivery details, items and the bill.

5. UI/UX Design | تصميم واجهة المستخدم

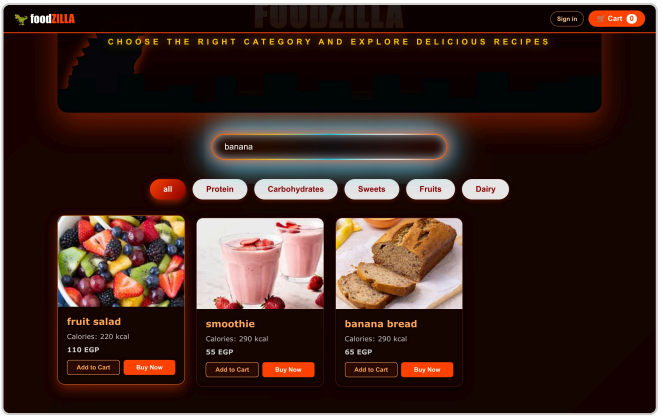
Screenshots | لقطات الشاشة



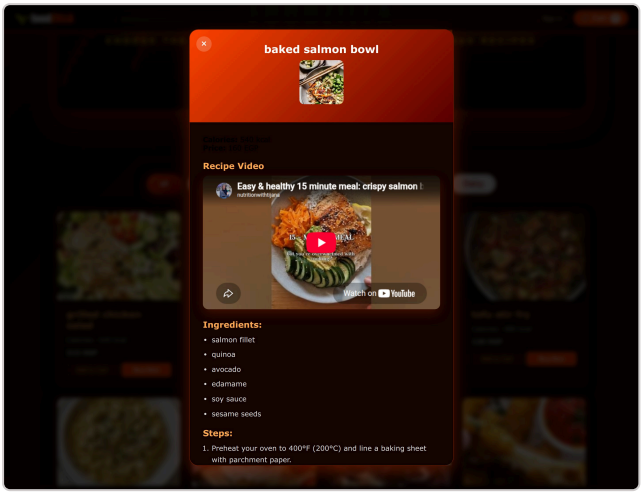
Protein group



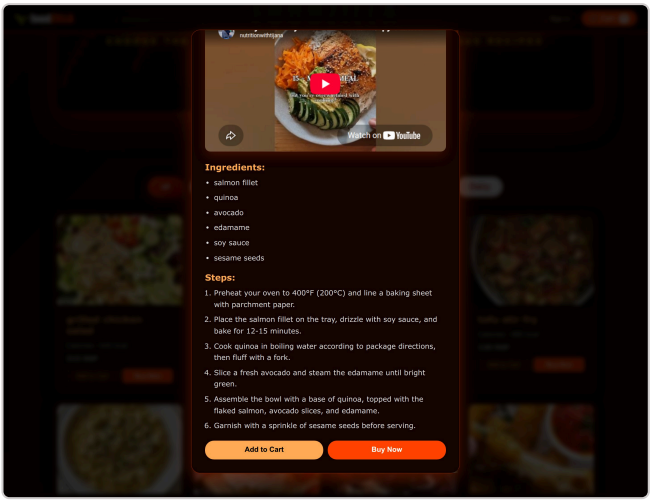
Search ”banana”



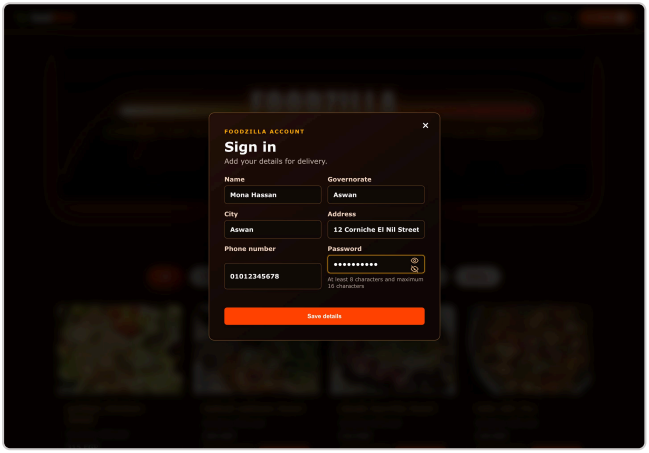
Recipe popup



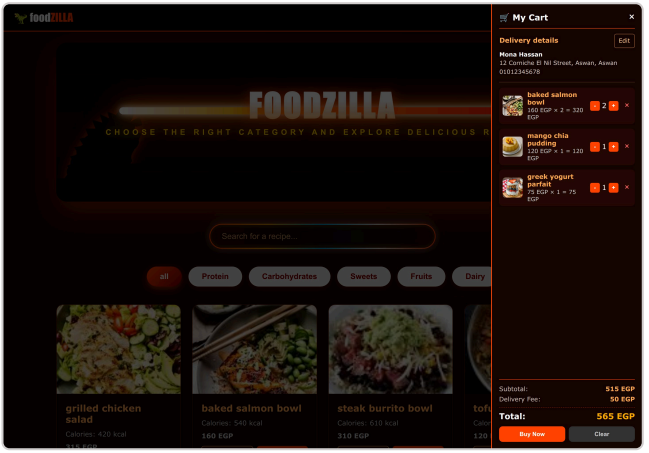
Steps and buttons

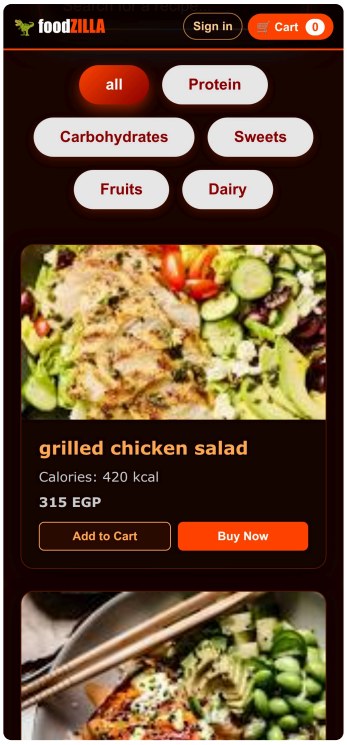


Sign in



My Cart





Try the app: https://mr-basem.github.io/Foodzilla/

Design System | نظام التصميم

Element	Current implementation
Logo	"🦖 food**ZILLA**": "food" in white and "ZILLA" in fire orange, in the navbar. The header shows a Godzilla silhouette drawn as an SVG. The poster and slides redraw that silhouette with its fire beam (`docs/assets/mark.svg`).
Primary color	Fire orange `#FF4500`, used for buttons, borders, glows and "ZILLA"
Accent colors	Beam yellow `#FFAE00`, warm orange `#FFAA54` for headings in cards, and gold `#FFCC00` for the header subtitle
Background	A slowly moving dark red gradient, `#2B0000` → `#1A0500` → `#0D0200`
Title font	Impact (with Arial Black as backup), uppercase, with a white-to-orange gradient. The poster and slides use Anton, its Google Fonts twin.
Body font	Segoe UI, Tahoma, Geneva, Verdana
Cards	Dark, rounded corners, an orange glow on hover, photo on top
Motion	Five CSS animations: fire background, lightning, monster rumble, fire beam and title shake

Element	Current implementation
Layout	Centered column up to 1200 px wide. Under 520 px the grid becomes one column.

6. Data Sources | مصادر البيانات

Data Sources | مصادر البيانات

Source	Type	Description
`project.js`	Written by the team	31 recipes with 163 ingredients and 158 steps in all. Prices from 20 to 400 EGP, calories from 170 to 720 kcal.
Recipe photos	Image files in the repo	31 `.jpg` files named like `download (40).jpg`, one per recipe.
YouTube	Embedded videos	31 cooking videos, one per recipe, linked by `youtubeUrl`.
Sign in form	User input	Name, governorate, city, address and phone, saved in `localStorage`.

Recipes | الوصفات

Group	Count	Recipes
Protein	4	grilled chicken salad, baked salmon bowl, steak burrito bowl, tofu stir fry
Carbohydrates	7	beshamel pasta, spagetti, donut, Spongebob burger, pizza, fried chicken, garlic bread
Sweets	8	chocolate cake, pancake, sweet crepes, golden balls fried sweets, cookie, fudge brownies, cinnamon rolls, apple pie
Fruits	7	fruit salad, cocktail, smoothie, banana bread, fruit curry, mango chia pudding, strawberry sorbet
Dairy	5	mac and cheese, greek yogurt parfait, cheese fondue, mozzarella sticks, creamy alfredo sauce

Data Model | شكل البيانات

There is no database. The app keeps three kinds of data:

Data	Where it lives	Fields	Lasts
Recipe	`project.js`, in `recipes.protein`, `.carbs`, `.sweets`, `.fruits`, `.dairy`	`name`, `icon`, `price`, `ingredients`, `calories`, `youtubeUrl`, `steps`	Until the file is edited
Cart item	The `cart` list in `app.js`	`name`, `icon`, `price`, `qty`	Until the page reloads

Data	Where it lives	Fields	Lasts
Customer	`localStorage` key `foodzillaCustomer`	`name`, `governorate`, `city`, `address`, `phone`	In this browser until site data is cleared

The bill is worked out every time the cart changes: subtotal = each price × its quantity, total = subtotal + 50 EGP delivery (`DELIVERY_FEE`). A recipe with no price would cost 100 EGP (`DEFAULT_RECIPE_PRICE`).

External APIs | الخدمات الخارجية

API	Purpose	Rate Limit
YouTube embed	Plays the recipe video inside the popup	None for normal viewing. Needs internet.

Limits | حدود المشروع

These are the shortcuts and gaps in the current version:

- Orders are a demo. "Buy Now" shows a thank-you message, but the order is not sent to anyone and no money is taken.
- The password on the Sign in form is checked for length and then thrown away. There are no real accounts.
- Delivery details are saved in plain text in this one browser, and the cart is lost when the page reloads.
- The phone box accepts 10 or 11 digits on Sign in, but the Edit form in the cart accepts any text.
- Several recipes in the groups are not light meals (for example pizza, donut and fried chicken), and calories are estimates written by the team.
- The recipe videos need internet. The photos and everything else work offline.
- On a narrow phone the shaking "FOODZILLA" title is cut at the right edge.
- In the recipe popup, the "Calories" and "Price" lines are dark text on a dark background.

7. Educational Content | المحتوى التعليمي

Learning Objectives | أهداف التعلم

Through building this project, a student can practice:

1. Building a whole page with HTML, and showing or hiding parts of it (popup, panel, drawer) by adding and removing a CSS class.
2. Making motion with CSS alone: `@keyframes`, gradients, `clip-path` and glow shadows.
3. Storing data in a JavaScript object of lists, and drawing it on the page with a loop.

4. Filtering a list: by category, and by matching text in a name or in any ingredient.
5. Keeping a cart in memory, doing the bill math, and saving details with `localStorage`.

Cultural / Historical Context | السياق الثقافي

Topic	Description
Godzilla	Godzilla first appeared in a Japanese film in 1954 and became one of the most famous movie monsters in the world. Its atomic breath inspired the fire beam in the header.
Nutrition groups	Proteins, carbohydrates, dairy, fruits and sweets are the groups many school nutrition lessons use to explain a balanced plate. foodZILLA uses them as its five buttons.
Food in Egypt	Prices are in Egyptian pounds, and the delivery form asks for a governorate and city, the way addresses are written in Egypt.

Curriculum Alignment | الارتباط بالمنهج

Skill Area	What the Student Practices
Problem Solving	Turning "healthy food is boring" into a website people want to explore.
Computational Thinking	Splitting the app into data (`project.js`), look (`style.css`) and logic (`app.js`).
Creativity	Designing a movie-monster theme with fire, lightning and a skyline, using code only.
Collaboration	Sharing the code on GitHub and publishing it with GitHub Pages.
Technical Writing	Writing 158 recipe steps and the project README in clear, short sentences.

8. Marketing Plan | خطة تسويقية

Target Audience | الجمهور المستهدف

- **Primary:** teens and young adults aged 13 to 25 who like games and movies and think healthy food is boring.
- **Secondary:** university students on a budget, gym-goers who watch protein and calories, and parents who cook with their children.

Platforms | المنصات

- Phone and laptop browsers. The site works on both and needs no install.
- Short cooking clips on Instagram, TikTok and YouTube Shorts with a "find it on foodZILLA" link, plus a QR code on posters at schools, gyms and youth forums.

9. Student Worksheet | ورقة عمل الطالب

My Project Information | معلومات مشروعي

Student Name: Abdel Wahab
Project Title: foodZILLA: The Beast of Healthy Recipes

What is your project about? | عن ماذا يتحدث مشروعك؟

foodZILLA is a website with 31 recipes in five nutrition groups. Every recipe has a video, ingredients, steps, calories and a price, and you can put meals in a cart.

What problem does it solve? | ما المشكلة التي يحلها؟

It shows young people that healthy food can be exciting, and puts everything they need to cook a meal on one page.

My Data | بياناتي

Where does data come from?

The team wrote the 31 recipes in `project.js` and picked a photo and a YouTube video for each one. Visitors type their delivery details into the Sign in form.

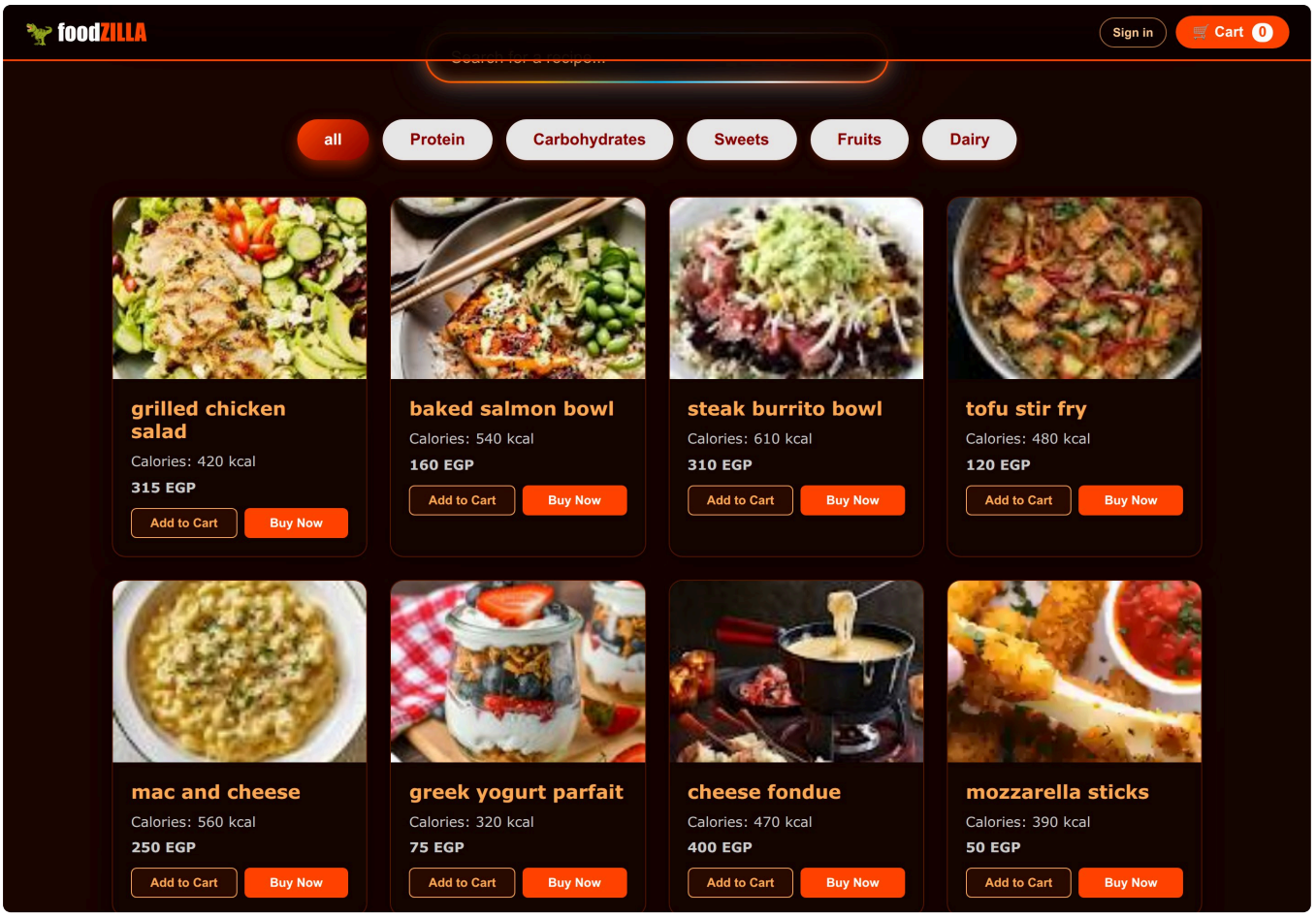
What data is stored?

The recipes are stored in the code. The cart is kept only while the page is open. The visitor's name, governorate, city, address and phone are saved in their own browser.

What I Learned | ماذا تعلمت

1. **The hardest part:** lining up the fire beam with Godzilla's mouth and timing it with the lightning and the shaking title.
2. **The most fun part:** watching the header come alive the first time all the animations ran together.
3. **If I had more time:** I would send real orders to a server and add a nutrition score to every recipe.
4. **One thing I would do differently:** choose lighter recipes for every group, so that each one is truly healthy.

Screenshot of My Project | لقطة من مشروعي



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foodZILLA team · AFFIAT · Afro-Asian Forum for Innovation and Technology · Aswan Finals 2026